

② (a) Explain how Schrodinger equation helps to understand the quantization of energy levels in 1D potential well. [4M]

(b) Apply the normalization condition to obtain wavefunctions for particle in 1D pot well and plot them [3M]

(c) An electron possesses a wavelength of $2 \text{ \AA}$ in the ground state. Calculate the length of the potential well. [1M]

Hint (1):
$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - \underline{V(x)}) \psi(x) = 0$$

$\Rightarrow$ Inside the well,
$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m E}{\hbar^2} \psi(x) = 0$$



$$\Rightarrow \frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2} \psi(x)$$

$$\text{Let } k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow E_n = \frac{n^2 \hbar^2}{8ma^2}$$

we have energy also quantized.

B.C. (ii) At $x=a$,

$$\psi(x=a) = 0$$

$$A \sin ka = 0$$

$$\Rightarrow \sin ka = 0$$

$$\Rightarrow ka = n\pi,$$

where $n=1, 2, 3, \dots$

Solutions of (A) are of the form

$$\psi(x) = A \sin kx + B \cos kx \dots$$

$$\Rightarrow k_n = \frac{n\pi}{a}$$

B.C. (i) At $x=0$, $\psi(x=0)=0 \Rightarrow B=0$

$$\Rightarrow \psi(x) = A \sin kx$$

Since k is quantized,

$$\Rightarrow E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2} \text{ and } k_n^2 = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{\hbar^2 k_n^2}{2m}$$

(b) Wavefunction of quantum number n is

$$\psi_n(x) = A \sin K_n x = A \sin \frac{n\pi}{a} x$$

Probability density is given by $|\psi(x)|^2$

so that total probability is

$$\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1 \quad \Leftarrow \text{Normalization condition}$$

$$\left(\int \cos 2\theta dx = \frac{\sin 2\theta}{2} \right)$$

But for 1D potential well

$$\Leftrightarrow \frac{A^2}{2} x \Big|_0^a - \frac{A^2}{2 \cdot 2a} \sin^2 \frac{2\pi x}{a} \Big|_0^a = 1$$

$$\Leftrightarrow \frac{A^2}{2} a - 0 = 1$$

$$\Leftrightarrow A^2 = \frac{2}{a} \Leftrightarrow A = \sqrt{\frac{2}{a}}$$

For $n=1$, Let $n=10$

Let $n=10$

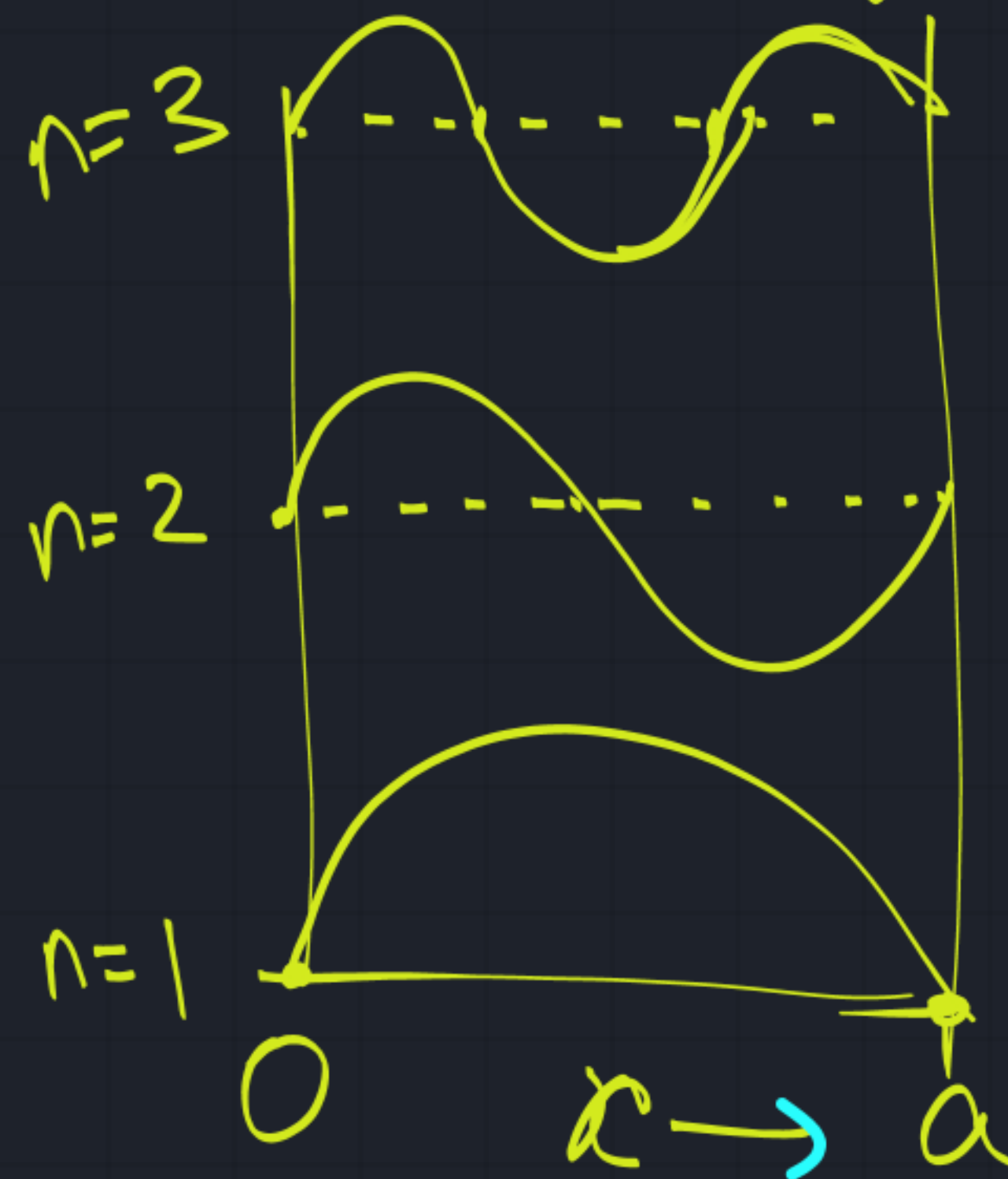
$$\int_0^a A^2 \sin^2 \frac{n\pi x}{a} dx = 1$$

$$\Rightarrow \psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$\Leftrightarrow \int_0^a \frac{A^2}{2} (1 - \cos \frac{2n\pi x}{a}) dx = 1 \quad \left(\cos 2x = 1 - 2\sin^2 x \right)$$

$$\Rightarrow \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \boxed{\frac{n\pi x}{a}}$$



$$\psi_1(x) = \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a}$$

$$\psi_2(x) = \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a}$$

$$\psi_3(x) = \sqrt{\frac{2}{a}} \sin \frac{3\pi x}{a}$$

(c) $\lambda_1 = 2\text{\AA}$

$$\hookrightarrow \boxed{k_1 = \frac{2\pi}{\lambda_1}}$$

$$k_1 = \frac{\pi}{a}$$

$$\hookrightarrow \frac{\pi}{a} = \frac{2\pi}{\lambda_1}$$

$$\therefore a = \frac{\lambda_1 \cdot \pi}{2\pi} = \frac{\lambda_1}{2}$$

$$\boxed{\therefore a = 1\text{\AA}}$$

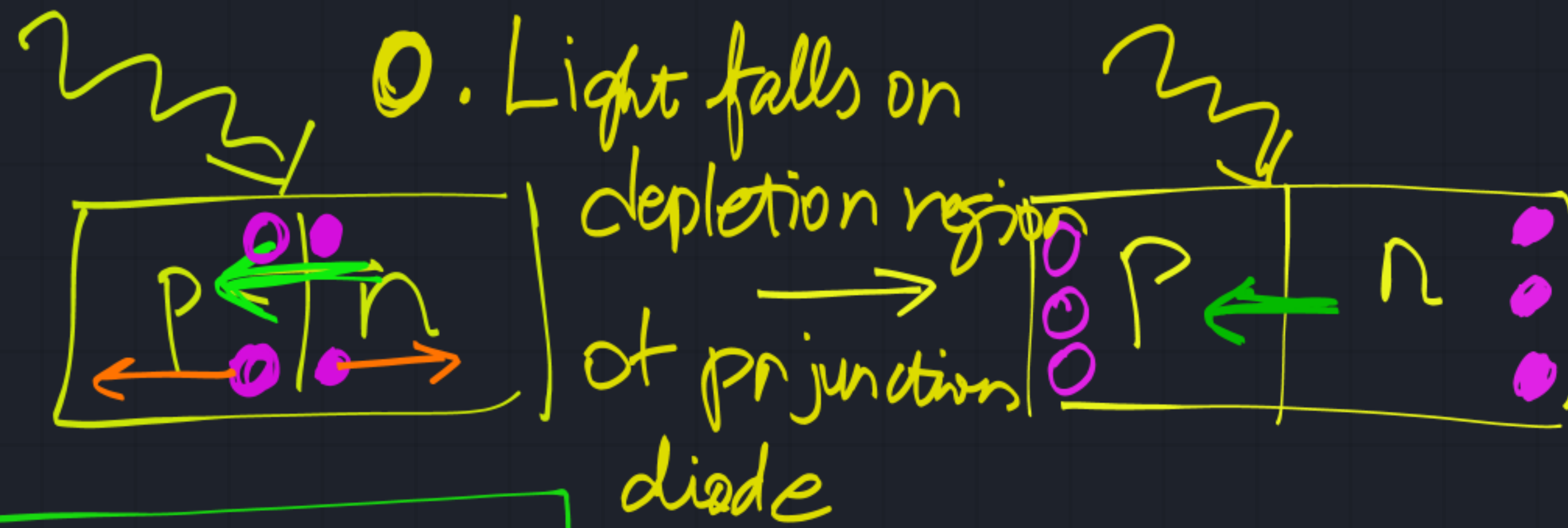
③ (a) Explain photovoltaic effect. [1M]

(b) Elucidate role of depletion region in p-n junction diode in conversion of
 clear ← light power into electrical power in the photovoltaic cell. [4M]

(c) Draw I-V characteristics of solar cell and write the expression for
 f.f. and η . [3M]

(a) Conversion of light energy into electrical energy inside a
 p-n-junction diode under zero bias is called photovoltaic effect.

Application	Working principle	Conversion of energy		Bias
		from	to	
Solar cell	photovoltaic effect	light	electrical	Zero
LED	Electroluminescence	electrical	light	Forward
Photodiode	Photoconductive effect	light	electrical	Reverse

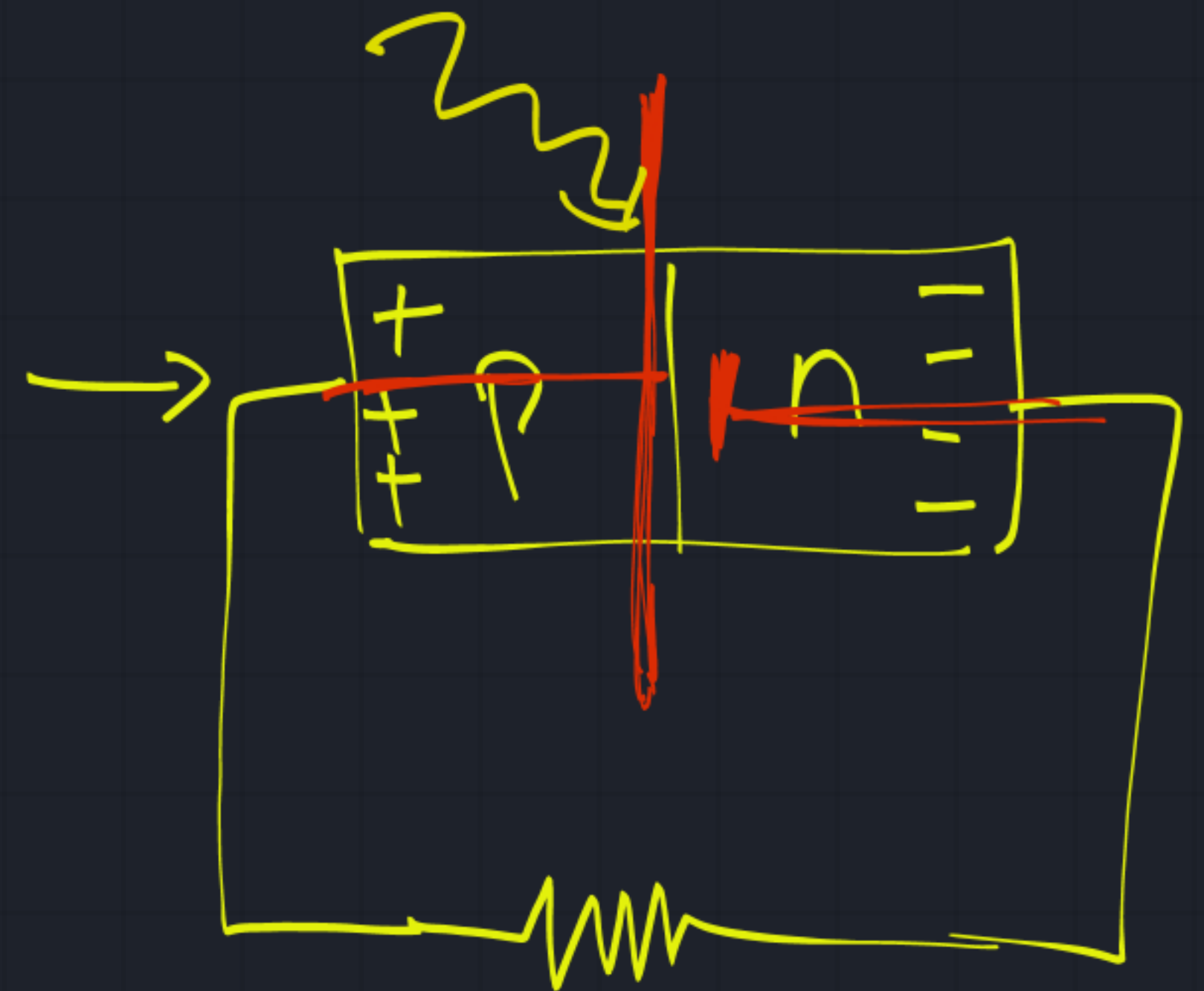


$$F_{\text{electron}} = q E$$

$$F_{\text{hole}} = -e E$$

1. Generation of excess electron-hole pairs.

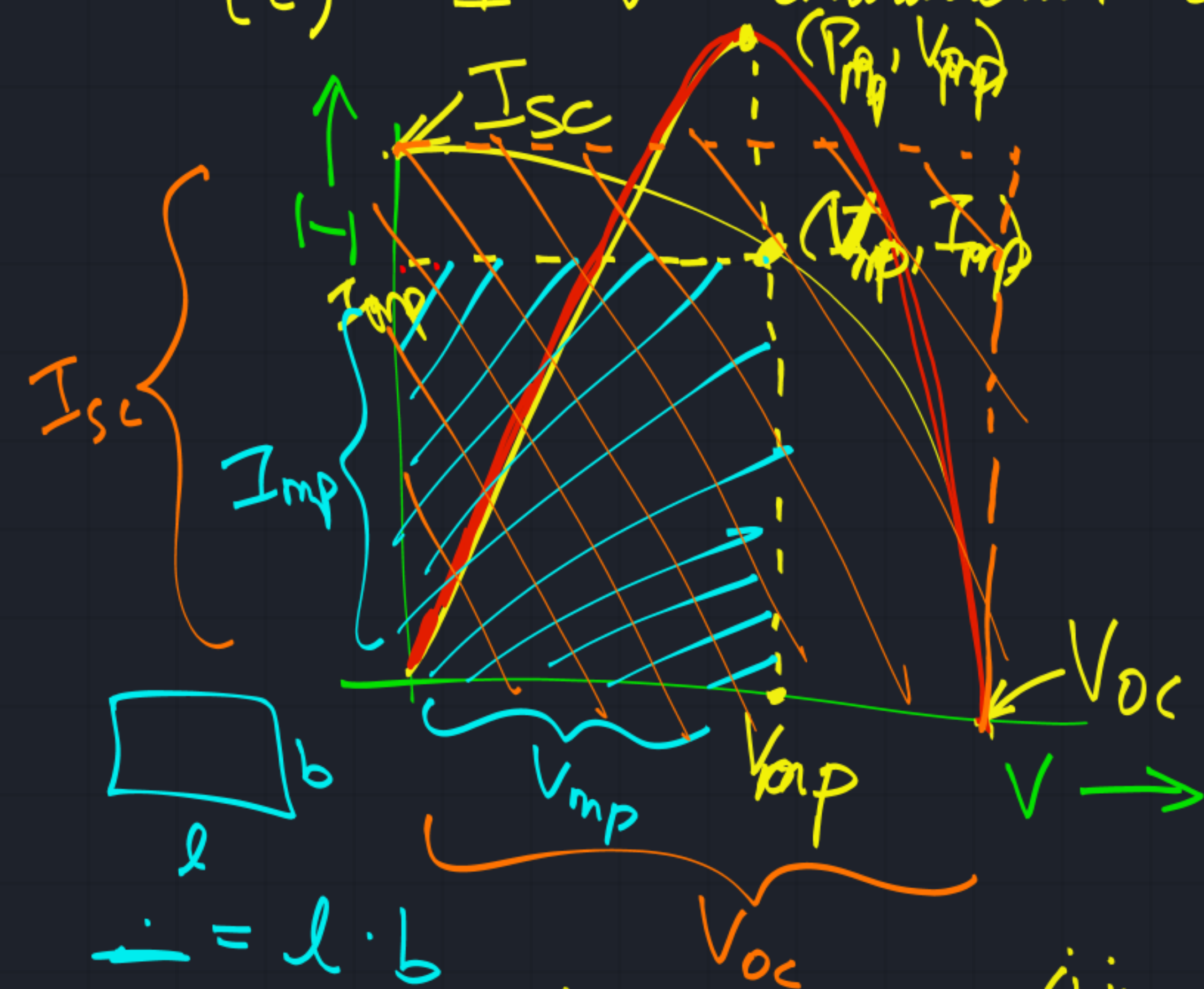
3. Separation of excess electron-hole pairs due to built-in electric field



4. Generation of photo-voltage.

2.

(c) I-V characteristics of solar cell.



V_{oc} = Open circuit voltage

I_{sc} = short circuit current

V_{mp} = voltage at max. power output

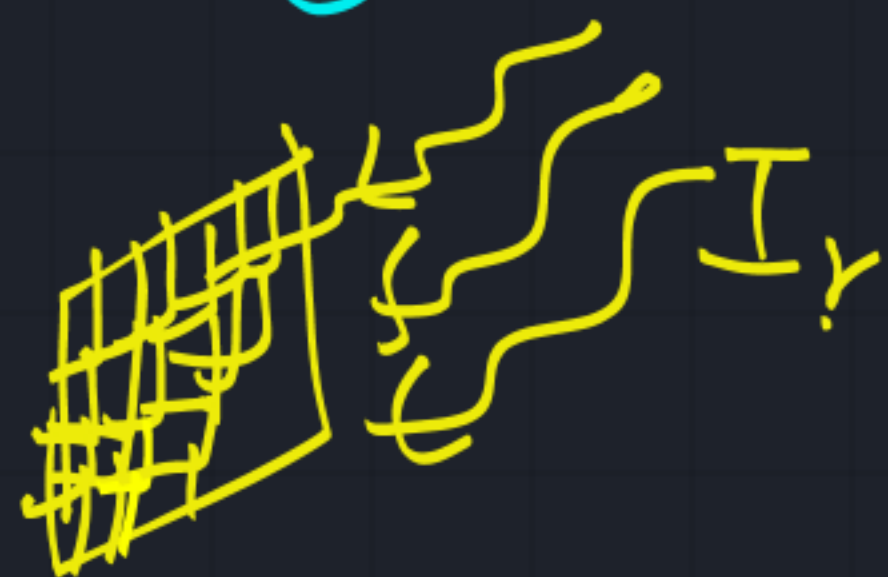
I_{mp} = current at max power. output

$$(i) F.F. = \frac{\text{max output power}}{\text{ideal output power}} = \frac{V_{mp} \cdot I_{mp}}{V_{oc} \cdot I_{sc}}$$

where F.F. is Fill-factor

$$(ii) \eta = \frac{\text{Output power}}{\text{Input power}} \times 100 = \frac{V_{mp} \cdot I_{mp}}{I_r \cdot A_{\text{panel}}} \times 100 \leq 1$$

where I_r = Input irradiance, A_{panel} = area of panel



① (a) Explain significance of Heisenberg's uncertainty principle with one suitable example. [4M]

(b) Describe how H. unc. principle helps to understand the non-existence of electron inside the nucleus of atom. [3M]

(c) An electron is confined inside a box of length 10^{-9}m . Calculate minimum uncertainty in the momentum. [1M]

Heisenberg uncertainty principle:

Example: Cricket ball of mass = 250g = 0.25kg
Velocity of ball = 140 kmph
 $\Delta p = 0.25\text{kg} \cdot 40\text{ms}^{-1} = 10\text{kgms}^{-1}$

We cannot measure the position and momentum of a particle simultaneously (or at the same time)

Mathematically, H. u. p. states that $\Delta x \cdot \Delta p \geq \frac{h}{2}$,

where Δx is uncertainty in position, Δp is uncertainty in momentum. $\Rightarrow$

$$m = 0.25 \text{ kg}$$

$$(9.1 \cdot 10^{-31} \text{ kg})$$

$$v = 40 \text{ ms}^{-1}$$

$$\Delta p = 0.25 \cdot 40 \text{ kg ms}^{-1} = 10 \text{ kg ms}^{-1}$$

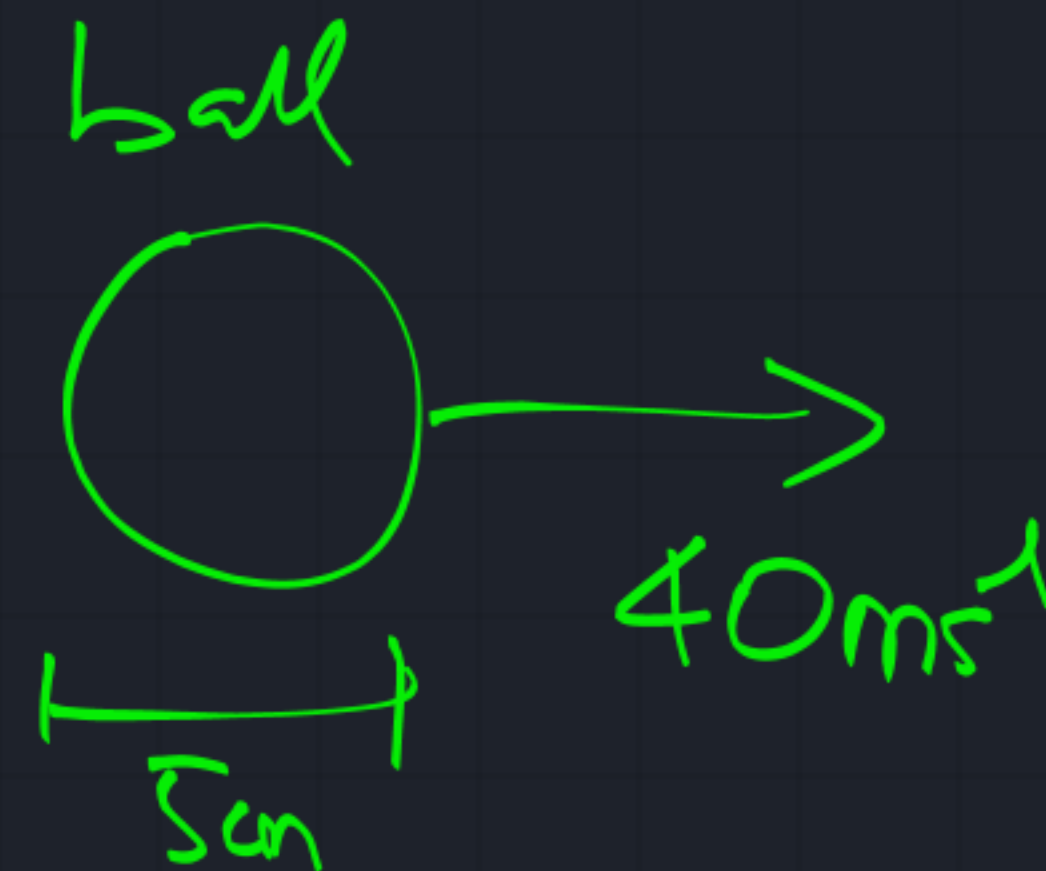
$$\text{Then } \Delta x \geq \frac{\hbar}{2 \cdot \Delta p} = \frac{h}{4\pi \cdot \Delta p}$$

$$= \frac{6.625 \cdot 10^{-34}}{4 \cdot 3.14 \cdot 10}$$

$$= 0.5 \cdot 10^{-35}$$

$$= 5 \cdot 10^{-36} \text{ m}$$

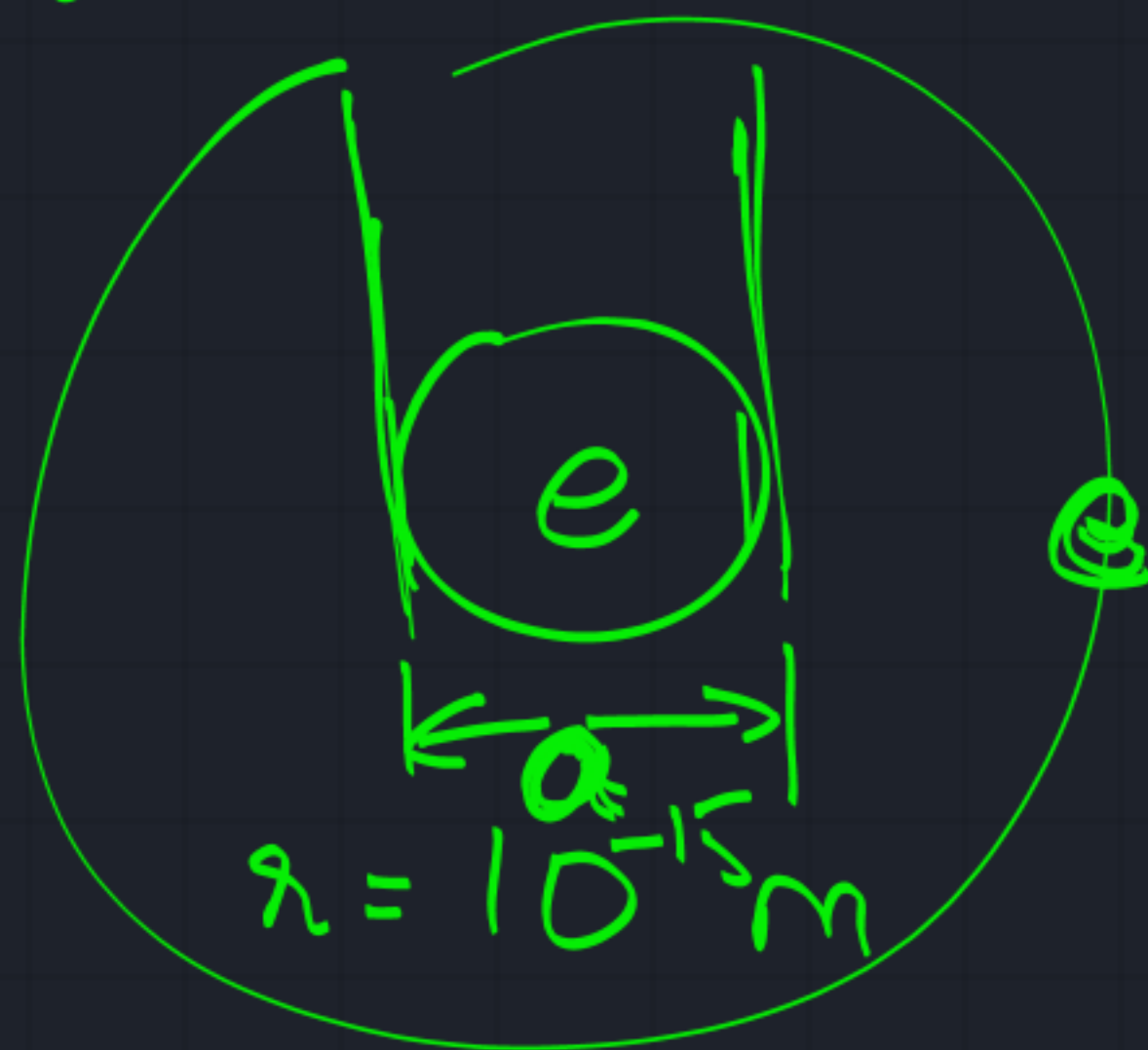
$$= 5 \cdot 10^{-36} \text{ m}$$



∴
electron

(b) Electron in a nucleus : Proof by contradiction.

Let us Assume that electron can stay inside a nucleus.



a = width of potential well

r = radius of nucleus = 10^{-15} m

$$a = 2 \cdot r = 2 \cdot 10^{-15} \text{ m}$$

$$E_n = \frac{n^2 h^2}{8ma^2} \Rightarrow E_1 = \frac{h^2}{8ma^2} = \frac{h^2}{32 \cdot m r^2} = \frac{(6.625 \cdot 10^{-34})^2}{32 \cdot 9.1 \cdot 10^{-31} \cdot 10^{-30}}$$

(10¹⁰ eV) This energy is not physical, $E_1(\text{eV}) = \frac{6.625 \cdot 6.625 \cdot 10^{-68}}{32 \cdot 9.1 \cdot 10^{-31} \cdot 1.6 \cdot 10^{-19}} = 10^{10} \text{ eV}$

$\therefore$ By the method of proof by contradiction, electron cannot stay inside a nucleus

$$(c) \Delta x = 10^{-9} \text{ m}$$

$$\Delta p = ?$$

By H.U.P.,

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

$$\Delta p \geq \frac{h}{2 \Delta x} = \frac{h}{4\pi \Delta x}$$

$$= \frac{6.625 \cdot 10^{-34}}{4.314 \cdot 10^{-9}}$$

$$= \frac{1.54 \cdot 10^{-25}}{2} = 0.5 \cdot 10^{-25} = 5 \cdot 10^{-26} \text{ kgms}^{-1}$$

④ (a) Describe how the direct band gap materials have superior performance than the indirect band gap materials in LEDs and LASERS.

(b) What are the essential conditions for an ordinary light to be called LASER?

(c) Briefly explain lasing action in semiconductor LASER with the help of a suitable band diagram.

⑤ (a) How does principles of Quantum Free Electron theory are applied to elucidate the electronic specific heat of metals to overcome failure of classical FET.

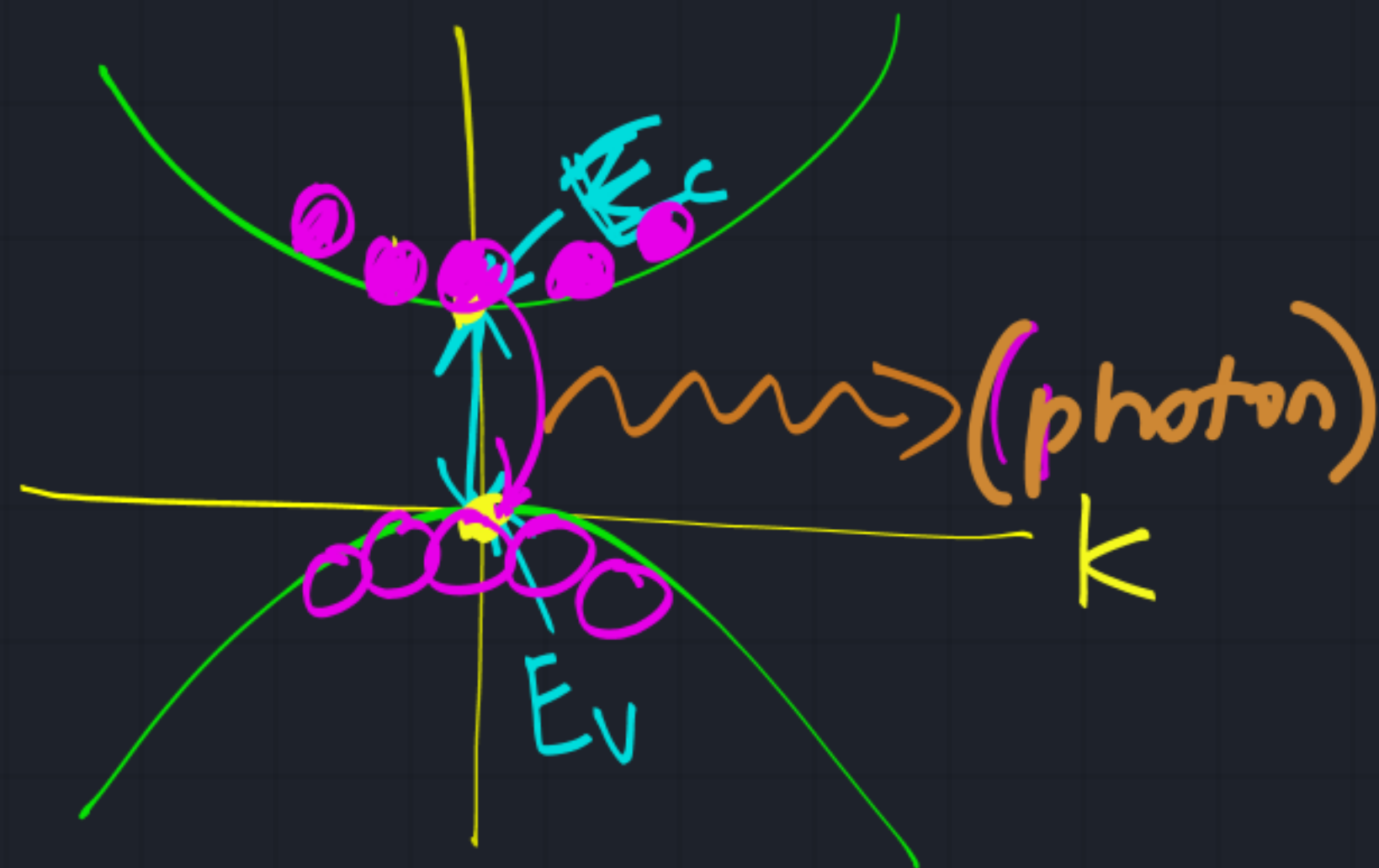
(b) What is F-D distribution function? Explain its temperature variation at $T=0K$ & $T>0K$

(c) Fermi level of silver is $5.5eV$. Calculate the probability of finding the electron at energy of $0.3eV$ above Fermi level at $1000K$.

④ (a) Direct band gap s.c.

compound

II	IV	V
Ga	Si	As
	Ge	P



$$(i) \quad k @ E_c = k @ E_v$$

(ii) electron + hole $\rightarrow$ photon
(3 particle process)
is efficient

Indirect band gap s.c.



$$k @ E_c \neq k @ E_v$$

electron + hole $\rightarrow$ photon + phonon
(4 particle process is NOT efficient)



- (c)

